

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Class : SYIT	Batch : 1
Date of Assignment : 15-07-2026	Date/Time of Submission : 15-07-2026

Queue Application:

- a. Write a program to: a. Implement a queue using an array.
code:-

```
import array as arr
que=[]
arr1=array("i",que)
def enqueue(data):
    print("Data inserted",data)
    arr1.append(data)

def dequeue():
    arr1.pop(0)
    print("data removed")

def show():
    print("---queue---")
    print(arr1, "\n")

enqueue(11)
enqueue(12)
enqueue(13)

dequeue()
show()
print("Panvir, S031")
```

Output:-

```
-----
Data inserted 11
Data inserted 12
Data inserted 13
data removed
---queue---
array('i', [12, 13])

Panvir, S031
```

- b. Simulate a simple queuing system (e.g., customer service queue).

Code:-

```

import time
que=[ ("Panvir",2), ("Rahul",3), ("Krish",4) ]
print ("---Bookings---")
print(que)

while que:
    job,timer=que.pop(0)
    print(f"Running: {job} {timer}|sec")
    time.sleep(timer)
print(que)
print("Panvir, S031")

```

Output:-

```

===== ]
---Bookings---
[('Panvir', 2), ('Rahul', 3), ('Krish', 4)]
Running: Panvir 2sec
Running: Rahul 3sec
Running: Krish 4sec
[]
Panvir, S031

```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?